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SwiftLog Logistics Analytics

Data Analytics Track | Data Preparation: Transformation Log

Key Takeaways

- **All 11 CSV files were imported and cleaned in Power Query,** producing an 11-table model (5-dimension tables and 6 fact tables). Delivery_Performance was merged into Shipments and is not loaded on its own.
- **The data is structurally clean:** every key is unique, no missing values were found, and there are no orphan links between tables. Cleaning therefore focused on types, text standardisation, dates and business fields.
- **Four preparation issues were handled and documented:** order dates stored as text, repeated inventory rows (stock movements, kept as advised by admin), duplicated delay information in two tables, and returns that link to orders but not products.
- **Every added field is defined in the data dictionary,** and all assumptions used during preparation are listed so results can be traced and challenged.

1. Approach and General Steps

All 11 SwiftLog CSV files were imported with Get Data and transformed in the Power Query Editor. The steps below were applied to every table before the table-specific steps in section 2. Screenshots of the Applied Steps pane for each query are kept in the Images folder.

Step	How it was done in Power Query	Purpose
Use first row as headers	Home, Use First Row as Headers	Column names come from the CSV header row.
Set data types	Click the type icon on each column (IDs and names as Text; counts as Whole Number; price as Fixed decimal number; OrderDate as Date using the English (United States) locale)	Correct types allow calculations and prevent date misreads such as 7/22/2026.
Clean text	Transform, Format, Trim and Clean; UPPERCASE on ID columns; Capitalize Each Word on names, cities, categories and statuses	Prevents mismatched keys and inconsistent labels.
Remove blank rows	Home, Remove Rows, Remove Blank Rows	Removes empty records. None were found.
Check column quality	View, Column quality and Column profile	Confirms 0% errors and 0% empty values in every column.
Remove duplicates on the key	Home, Remove Rows, Remove Duplicates on each table's ID (not applied to Inventory)	Guarantees one row per entity. No duplicate keys were found.

2. Transformation Log by Table

Each row lists the source file, the final model table, the steps applied, the reason and the check result.

#	Table (source to model)	Steps applied	Reason	Check and result
1	Customers → Dim_Customers	Promoted headers; types set to Text; trimmed and cleaned; CustomerID uppercased; City capitalised; City renamed CustomerCity; CustomerName split by the first space and the numeric part kept as CustomerNumber (Whole Number), the text part deleted; duplicates removed on CustomerID. DAX calculated columns added after loading: OrdersPlaced and OrderFrequencyBand.	Consistent keys and labels; a numeric customer number for sorting; order frequency analysis by customer.	1,000 rows; 1,000 unique IDs; 0 errors; 991 customers have orders and 9 have none.
2	Orders → Fact_Orders	Types set; OrderDate converted to Date using the English (United States) locale; IDs trimmed and uppercased; duplicates removed on OrderID; Status capitalised and renamed OrderStatus; date columns added (OrderYear, OrderMonth, OrderMonthName, OrderQuarter); conditional column IsDelivered (1 if OrderStatus is Delivered, otherwise 0).	Dates arrive as text in month/day/year format; time analysis and Delivered-only revenue need these columns.	5,000 rows; 0 date errors; dates run from 1 January to 8 September 2026; every order links to a valid customer.
3	Order_Items → Fact_OrderItems	Types set (Quantity as Whole Number); OrderID and ProductID trimmed and uppercased; duplicates removed on OrderItemID. DAX calculated columns added after relationships were built: LineRevenue, HasReturn, DeliveryStatus and ShipmentDelayDays.	Items carry no price, so revenue is quantity × unit price; delivery and return flags let product-level views work without two-way filters.	15,000 rows; 4,764 orders have items and 236 orders have none; total LineRevenue ties to the source calculation (21,668,066); 1,421 item lines belong to returned orders.
4	Products → Dim_Products	Types set (UnitPrice as Fixed decimal number); ProductID uppercased; Category capitalised; duplicates removed on ProductID; conditional column PriceBand added (under 150 Low; 150 to 350 Mid; over 350 High).	Price bands allow product pricing analysis.	250 products in 3 categories; 0 errors.
5	Shipments → Fact_Shipments	Types set; IDs trimmed and uppercased; duplicates removed on ShipmentID; merged with Delivery_Performance on ShipmentID (Left Outer); only Status expanded and renamed DeliveryStatus (capitalised); conditional columns IsDelayed and DelayBucket added.	Delivery status lives in a separate table; merging gives one shipment table and removes a duplicate DelayDays column.	5,000 rows; every shipment matched a delivery record; DelayDays is identical in both tables on all rows; On Time = 0 to 2 days and Delayed = 3 to 7 days with no overlap.
6	Delivery_Performance (staging)	Types set; text trimmed and cleaned; Enable load switched off after merging into Fact_Shipments.	Avoids a duplicate table in the model.	5,000 rows; used only as a merge source.
7	Returns → Fact_Returns	Types set; OrderID uppercased; Reason capitalised; duplicates removed on ReturnID.	Standard reason labels for return analysis.	500 returns across 473 orders; 3 return reasons.
8	Inventory → Fact_Inventory	Types set; WarehouseID and ProductID uppercased; duplicates not removed (admin: repeated rows are stock movements); Group By WarehouseID and ProductID with Sum of Stock as StockUnits; conditional column StockStatus added (under 100 Low; 100 to 399 Moderate; 400 and above Healthy).	One stock figure per warehouse and product for reporting, as advised by admin.	4,000 rows grouped to 2,455 warehouse-product pairs; total stock 1,007,764 equals the raw total.
9	Warehouses → Dim_Warehouses	Types set; WarehouseID uppercased; Location capitalised and renamed WarehouseCity; duplicates removed on WarehouseID.	Clear field name that is distinct from customer city.	15 warehouses (Lagos 8, Abuja 4, Kano 3).
10	Suppliers → Dim_Suppliers	Types set; text trimmed; SupplierID uppercased; duplicates removed on SupplierID. No relationships created.	Admin: suppliers are a standalone reference table.	150 suppliers; no link to products or shipments in the data.

#	Table (source to model)	Steps applied	Reason	Check and result
11	Employees → Fact_Employees	Types set; IDs uppercased; duplicates removed on EmployeeID.	Headcount per warehouse.	200 employees; every employee links to a valid warehouse.
12	Date table → Dim_Date	Created with DAX (CALENDAR, 1 January to 31 December 2026) with columns Year, MonthNumber, Month, Quarter and YearMonth; Month sorted by MonthNumber; marked as the date table; related to Fact_Orders[OrderDate].	Time intelligence and clean date slicers.	365 dates covering all order dates.

3. Data Quality Checks

Check	Method	Result
Missing values	Column quality and column profile on every table	None found in any of the 11 files.
Duplicate keys	Remove Duplicates on each table ID; compared row counts before and after	No duplicate IDs in any table.
Exact duplicate rows	Checked all columns on every table	Only Inventory has exact repeats: 3 rows (829 units, 0.08% of stock). Retained, as admin advised that repeated rows are stock movements.
Orphan links	Checked every foreign key against its parent table	0 orphans in all 10 key links (orders to customers, items to orders and products, shipments to orders and warehouses, delivery records to shipments, returns to orders, inventory to warehouses and products, employees to warehouses).
Date parsing	Date type with English (United States) locale	0 errors; orders from 1 January to 8 September 2026. September is a partial month (154 orders).
Delay information	Compared DelayDays in Shipments and Delivery_Performance	Identical on all 5,000 rows.
Orders without items	Compared order IDs with Order_Items	236 orders (4.7%) have no items, so they have no revenue. They remain in order counts.
Order status vs shipments	Cross-tab of order status and shipments	All 5,000 orders have a shipment, including 1,626 Cancelled and 1,716 Pending orders. 315 returned orders are Cancelled or Pending.
Inventory repeats	Counted rows per warehouse and product	1,057 of 2,455 pairs repeat (up to 7 rows); 9 rows have zero stock.

4. Assumptions Applied During Preparation

#	Topic	What was done	Why
P1	Delayed vs On Time	Delayed means 3 to 7 days late; On Time means 0 to 2 days.	The data shows no overlap between the two groups, so the definition is inferred directly from it.
P2	Inventory aggregation	Rows summed per warehouse and product; no rows removed.	Admin guidance; the file has no date to identify the latest record. Summing movements may overstate current stock.
P3	Revenue basis	LineRevenue = Quantity × UnitPrice (current list price). No discounts, costs or tax; currency not stated.	Order items carry no price.
P4	Product returns	A return is flagged on every item of the returned order (HasReturn).	Returns link to orders, not products, so product return rates are approximate.

#	Topic	What was done	Why
P5	Analyst-chosen bands	PriceBand (150 and 350), StockStatus (100 and 400) and OrderFrequencyBand (2, 5 and 8 orders) use thresholds chosen by the analyst.	No business thresholds were provided; different cut-offs would move records between bands.
P6	Order status	Order status is kept as a separate field and not used to filter shipments.	Cancelled and Pending orders still have shipments, so status does not gate shipping in the data.

5. Calculated Fields and Data Dictionary

Field	Table	Type	Definition	Created by
OrderYear, OrderMonth, OrderMonthName, OrderQuarter	Fact_Orders	Number / Text	Year, month number, month name and quarter of OrderDate.	Power Query (Add Column, Date)
IsDelivered	Fact_Orders	Whole Number	1 if OrderStatus is Delivered, otherwise 0.	Power Query (Conditional Column)
OrderStatus	Fact_Orders	Text	Status of the order (renamed from Status).	Power Query (Rename)
CustomerCity	Dim_Customers	Text	City of the customer (renamed from City); used as the delivery destination region.	Power Query (Rename)
CustomerNumber	Dim_Customers	Whole Number	Numeric part of CustomerName.	Power Query (Split Column)
OrdersPlaced	Dim_Customers	Whole Number	Number of orders placed by the customer.	DAX calculated column
OrderFrequencyBand	Dim_Customers	Text	No orders; Low (1 to 2); Medium (3 to 5); High (6 to 8); Very high (9 and above).	DAX calculated column
PriceBand	Dim_Products	Text	Low (under 150), Mid (150 to 350) or High (over 350) by UnitPrice.	Power Query (Conditional Column)
DeliveryStatus	Fact_Shipments	Text	On Time or Delayed, from Delivery_Performance.	Power Query (Merge Queries)
IsDelayed	Fact_Shipments	Whole Number	1 if DeliveryStatus is Delayed, otherwise 0.	Power Query (Conditional Column)
DelayBucket	Fact_Shipments	Text	0-2 days (On Time), 3-4 days or 5-7 days, by DelayDays.	Power Query (Conditional Column)
StockUnits	Fact_Inventory	Whole Number	Sum of Stock for each warehouse and product.	Power Query (Group By)
StockStatus	Fact_Inventory	Text	Low (under 100), Moderate (100 to 399) or Healthy (400 and above) by StockUnits.	Power Query (Conditional Column)
LineRevenue	Fact_OrderItems	Fixed decimal	Quantity × UnitPrice of the product.	DAX calculated column
HasReturn	Fact_OrderItems	True/False	True if the item's order has at least one return.	DAX calculated column
DeliveryStatus, ShipmentDelayDays	Fact_OrderItems	Text / Whole Number	Delivery status and delay days of the order's shipment.	DAX calculated column (LOOKUPVALUE)
WarehouseCity	Dim_Warehouses	Text	City where the warehouse is located (renamed from Location).	Power Query (Rename)

Field	Table	Type	Definition	Created by
Date, Year, MonthNumber, Month, Quarter, YearMonth	Dim_Date	Date / Number / Text	Calendar columns for 2026.	DAX calculated table

6. Data Model Relationships

The model follows a star schema: dimension tables (customers, products, warehouses, suppliers, date) describe the business, and fact tables (orders, order items, shipments, returns, inventory, employees) record events. All relationships are single direction from the one side to the many side. A screenshot of Model view is kept in the Images folder.

From (one side)	To (many side)	Type	Status
Dim_Customers[CustomerID]	Fact_Orders[CustomerID]	One to many	Active
Fact_Orders[OrderID]	Fact_OrderItems[OrderID]	One to many	Active
Fact_Orders[OrderID]	Fact_Shipments[OrderID]	One to one	Active
Fact_Orders[OrderID]	Fact_Returns[OrderID]	One to many	Active
Dim_Products[ProductID]	Fact_OrderItems[ProductID]	One to many	Active
Dim_Products[ProductID]	Fact_Inventory[ProductID]	One to many	Active
Dim_Warehouses[WarehouseID]	Fact_Inventory[WarehouseID]	One to many	Active
Dim_Warehouses[WarehouseID]	Fact_Shipments[WarehouseID]	One to many	Active
Dim_Warehouses[WarehouseID]	Fact_Employees[WarehouseID]	One to many	Active
Dim_Date[Date]	Fact_Orders[OrderDate]	One to many	Active
Dim_Suppliers	(none)	Standalone reference table	No relationship, as directed by the admin

7. Optional Steps Not Applied

- **Warehouse label (merged column):** not created. WarehouseID and WarehouseCity are used directly in visuals.
- **Movement count in Fact_Inventory:** not created. Only the summed StockUnits is needed. The raw file has 4,000 movement rows and 2,455 warehouse-product pairs.
- **Assumed supplier link:** not created. Suppliers remain a standalone reference table, so no supplier-to-product assumption is applied in the model.

Recommendations

Improvements to data collection that would remove the assumptions above:

- **Add a supplier ID to the Products table** so supplier performance can be measured instead of discussed conceptually.
- **Add warehouse capacity** so utilisation can be calculated rather than estimated.
- **Add dispatch and delivery timestamps** so delivery time can be measured directly.
- **Date-stamp every inventory record** so the current stock level can be identified.
- **Record the product on each return** so product return rates are exact.

- **Align order status with shipments and returns** so Cancelled and Pending orders do not appear shipped or returned.